INDEPENDENT PRODUCT DISCOVERY & SERVICE DESIGN

MissionReady

Helping aerospace engineers become independently effective, sooner

A concept for improving engineer onboarding and trusted knowledge access in a technically complex, high-growth aerospace organization.

Focus	Engineer onboarding • Knowledge discovery • Operational readiness
Methods	Jobs-to-be-Done • Design Thinking • Outcome-Driven Innovation
Status	Independent concept; discovery and outcomes not yet validated

JOIN	FIND	VERIFY	PROGRESS
Understand the role	Locate trusted sources	Confirm version & access	Act or escalate safely

Portfolio integrity note
This case study uses illustrative hypotheses and proposed methods. It does not represent a commissioned engagement, validated customer research, a deployed product, or measured results.

01 / DEFINE THE OPPORTUNITY

Start with the engineer’s job—not the tool

In a growing aerospace organization, onboarding can involve much more than HR administration. Engineers may need to learn team responsibilities, locate current approved documentation, navigate internal processes, obtain system access, and identify the right people to consult.

Problem to investigate
How might we help newly hired engineers navigate unfamiliar organizational processes and find trustworthy, relevant information without adding administrative friction or compromising engineering controls?

> **PRIMARY JOB STATEMENT**
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> **“When I join a complex engineering organization, help me become independently effective in my role by giving me timely access to trusted information, clear next steps, and the right people to contact.”**

What is known—and what is not

TYPE	CURRENT POSITION
Working hypothesis	Information may be fragmented across systems, documents, and teams.
Working hypothesis	Process ownership and escalation routes may be unclear to new joiners.
Working hypothesis	Managers and mentors may spend time answering recurring procedural questions.
To validate	Which tasks create the most friction, for whom, and at what point in onboarding.
To validate	Whether the best response is a product, improved documentation, clearer ownership, or process redesign.

Discovery plan

- **Listen:** Interview recently hired engineers, engineering managers or mentors, and People Ops / IT / knowledge-governance partners.
- **Reconstruct:** Ask participants to walk through a recent onboarding task, including searches, handoffs, waits, and workarounds.
- **Map:** Document the tools, people, decisions, approvals, and points of uncertainty involved.
- **Synthesize:** Separate direct observations from interpretations; compare patterns across roles and access requirements.

Initial sample to explore: 5–8 recently hired engineers, 3–5 managers or mentors, and 2–3 support stakeholders. These are planning ranges, not a representative sample. Research would use authorized, non-sensitive information.

02 / MAP & PRIORITIZE

Find the unmet outcome

The Universal Job Map helps examine the whole job rather than optimizing a single screen. The steps below are a starting framework; research must establish how they appear in the real onboarding journey.

JOB STEP	ENGINEER'S QUESTION	OPPORTUNITY TO INVESTIGATE
1. Define	What do I need to accomplish?	Clarify expected outcome and constraints.
2. Locate	Where can I find the information or resources?	Improve discoverability and ownership visibility.
3. Prepare	What access, tools, approvals, or inputs are needed?	Surface prerequisites and dependencies.
4. Confirm	Is this information current, trusted, and authorized?	Make source, version, status, and permissions clear.
5. Execute	What steps should I follow?	Link to approved guidance and clear handoffs.
6. Monitor	Am I progressing? What is blocking me?	Show incomplete actions and dependencies.
7. Modify	What if circumstances change?	Route to updated information or human support.
8. Conclude	How do I know the task is complete?	Clarify completion criteria and next action.

Outcome-Driven Innovation (ODI)

ODI prioritizes desired outcomes by assessing importance and current satisfaction. Candidate outcome statements should be measurable from the engineer's perspective—for example, minimizing the time needed to locate the current approved document, or minimizing the time needed to identify the owner of an unfamiliar process.

Opportunity score = Importance + max(Importance − Satisfaction, 0)

ILLUSTRATIVE OUTCOME	IMPORTANCE	SATISFACTION	OPPORTUNITY
Find current approved document	9	3	**15**
Identify process owner	8	4	**12**
Understand next onboarding action	7	6	**8**

Important: Scores above are hypothetical examples to demonstrate the method—not survey data or findings. Actual scores would be collected from a suitable participant sample and interpreted alongside qualitative evidence.

03 / CONCEPT & VALIDATION

Design for trust, then test the value

If discovery validates the hypotheses around fragmented knowledge and unclear next steps, MissionReady could be explored as one possible response. The concept is intentionally framed as a hypothesis, not a predetermined solution.

PROPOSED CAPABILITY	USER VALUE	DESIGN GUARDRAIL
Personalized onboarding path	Role-relevant tasks, prerequisites, owners, and completion criteria.	Only show tasks appropriate to the user's role and permissions.
Trusted knowledge discovery	Find authorized sources with links, owner, version, and status where available.	Respect source-system permissions; do not create new access rights.
Guided next steps	Understand dependencies and when approval or access is required.	Do not replace formal engineering procedures or approvals.
Human escalation	Reach the appropriate expert when information is missing or unclear.	Abstain from unsupported answers and surface uncertainty.
Manager visibility	See onboarding progress and unresolved blockers.	Use proportionate, transparent reporting—not employee surveillance.

Security and governance by design

- Use approved environments and authorized data sources only; begin prototypes with synthetic or non-sensitive information.
- Enforce least-privilege access and the permissions of connected source systems.
- Show source links and document status where available; route conflicts to the designated authority.
- Do not present ungrounded AI output as authoritative; abstain or escalate when evidence is insufficient.
- Involve relevant security, engineering, legal, privacy, export-control, and information-governance stakeholders before any pilot, as applicable.

Six-week discovery and test plan

WEEK	FOCUS	OUTPUT
1	Align & scope	Target segment, research questions, boundaries, baseline plan.
2	Discover	Interviews and concrete onboarding task examples.
3	Synthesize	Evidence-based journey map, job map, prioritized outcomes.
4	Design	Low-fidelity prototype using synthetic or approved data.
5	Test	Task-based usability sessions and safety/accuracy scenarios.
6	Evaluate	Evidence summary and iterate / pilot / stop recommendation.

04 / MEASURE & LEARN

Define evidence before claiming impact

The pilot should begin with baseline measurement and agreed success criteria. Metrics below are candidates; targets should be set only after baseline collection and stakeholder alignment.

METRIC	HOW TO MEASURE	WHAT IT TELLS US
Time to locate approved document	Median time from task prompt to correct, authorized source.	Knowledge discoverability.
First-time task success	Share completing a defined task correctly without avoidable retries.	Task effectiveness.
Unresolved onboarding blockers	Count and age of blockers in the agreed scope.	Friction and dependencies.
Confidence in information	Post-task rating plus qualitative explanation.	Whether source and status cues support trust.
Mentor time on recurring questions	Self-reported or sampled time for defined repeat questions.	Potential capacity benefits.
Unsafe / unsupported answer rate	Test cases where guidance is ungrounded or fails to escalate.	Critical trust and safety performance.

Decision framework

- **Progress** if users complete selected tasks more reliably and can verify the authoritative source.
- **Protect** if access controls, approval paths, and appropriate escalation remain intact.
- **Reconsider** if the benefit does not justify integration, maintenance, governance, and adoption costs.
- **Choose the simplest effective intervention:** the answer may be better documentation, clearer ownership, or process redesign—not a new AI product.

Reflection

The value of this concept is the discipline of the approach: start with the job, investigate the current experience, identify unmet outcomes, design against evidence, and define measurable tests before scaling. The key question is not whether an AI assistant can be built, but whether engineers have an important underserved need—and which safe, practical intervention best addresses it.

> **PORTFOLIO INTEGRITY**
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> This is an independent case study concept. It is not a commissioned engagement, validated customer research, a deployed product, or a report of measured business results. The job map, ODI scores, solution, metrics, and timeline are illustrative planning artifacts. Add findings and impact claims only after research or testing.

Methods: Jobs-to-be-Done • Universal Job Map • Outcome-Driven Innovation • Design Thinking
Portfolio themes: Product discovery • Service design • People Operations • Organizational effectiveness • Digital transformation • AI-enabled knowledge access • Responsible product design